Arista Access Node Failure SOP (WFD-ACC-48 - Whitefield POP) - Optical Transport Failure (ALARM_3352)

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_3352

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node WFD-ACC-48 at the Whitefield POP is a critical failure domain that necessitates prompt attention to prevent network disruptions. This failure domain occurs due to the inherent vulnerability of optical transceivers to mechanical stress, environmental degradation, or physical damage, which can cause a degradation in optical signal quality, leading to Rx/Tx power loss or complete SFP/XFP failure. The Access Node, being a critical component of the network infrastructure, relies heavily on the integrity of its optical transport mechanisms to ensure reliable data transmission. Any failure in this domain can have far-reaching consequences, including network congestion, packet loss, and service disruptions. Therefore, it is essential to employ a systematic approach to diagnose and resolve this failure, ensuring minimal downtime and maximum network availability.

Pre-requisite Checks

- Verify that the remote hands are dispatched to the Whitefield POP to facilitate on-site intervention, if required.

- Confirm the status of redundant links to ensure alternative paths are available for traffic rerouting.

- Ensure that the network operations center (NOC) is notified of the failure to coordinate efforts and facilitate remote support.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to environmental factors	High
Fiber cuts or physical damage	Medium
Rx/Tx power loss due to optical signal quality degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface | include optical, Expected Output: Optical signal quality metrics (e.g., rx_power, tx_power, optical_signal_to_noise_ratio), Description: This command isolates the optical signal quality and verifies the integrity of the SFP/XFP transceiver.

Command: show interface | include optical_signal_quality, Expected Output: Optical signal quality metrics (e.g., signal_degradation,
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```
optical_signal_quality), Description: This command provides detailed
information about the optical signal quality and helps identify
potential issues with the SFP/XFP transceiver or fiber link.

Command: show interface | include sfp_status, Expected Output: SFP/XFP
status (e.g., online, offline, degraded), Description: This command
verifies the status of the SFP/XFP transceiver and helps determine if
the failure is due to a physical issue or environmental factor.
```

Resolution Steps

Step 1: Step: Verify SFP/XFP status and optical signal quality, Command: show interface | include sfp_status optical_signal_quality, Description: Gather detailed information about the SFP/XFP status and optical signal quality to determine the root cause of the failure.

Step 2: Step: Replace SFP/XFP transceiver (if failed), Command: no interface , Description: Physically replace the failed SFP/XFP transceiver with a new one and reconfigure the interface.

Step 3: Step: Verify fiber link integrity (if physical damage suspected), Command: show interface | include optical_signal_quality, Description: Verify the integrity of the fiber link and check for any signs of physical damage or degradation.

Post-Check Verification

```
Command: show interface | include optical_signal_quality, Expected
Output: Optical signal quality metrics (e.g., signal_degradation,
optical_signal_quality), Description: Verify that the optical signal
quality has been restored to normal levels after the resolution steps.

Command: show interface | include sfp_status, Expected Output: SFP/XFP
status (e.g., online, offline, degraded), Description: Verify that the
SFP/XFP transceiver has been successfully replaced or repaired.
```

Roll-back Procedures

- Step: Restore original SFP/XFP transceiver (if replaced), Command: interface , Description: Physically restore the original SFP/XFP transceiver and reconfigure the interface.

- Step: Revert interface configuration (if modified), Command: no interface , Description: Revert the interface configuration to the original state.